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A Project Report on- “Hydrophobic Concrete”

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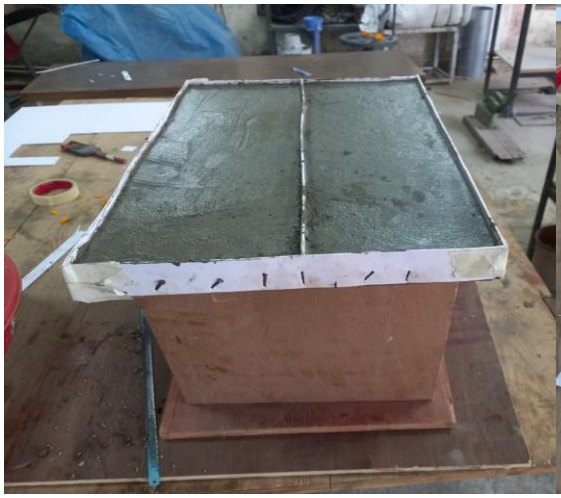

Acknowledgement

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Abstract

This report presents a comparative study of **hydrophobic concrete** and **Ordinary Portland Cement (OPC) concrete**, focusing on a model of a **slanted roof** where one half is cast with OPC concrete and the other half with hydrophobic concrete. The goal of this project was to explore the **water-repellent properties** of hydrophobic concrete and its advantages in construction applications. Through trials and adjustments, the mix design was refined, and challenges were overcome, ultimately leading to the successful creation of a hydrophobic concrete model. The report outlines the mix design, preparation method, curing process, and a discussion of the benefits and limitations of hydrophobic concrete, along with its real-world applications.

Introduction

In modern construction, water infiltration is a major concern, particularly in areas exposed to frequent rainfall. **Hydrophobic concrete** provides an innovative solution to this problem by incorporating water-repellent agents that prevent water from penetrating the structure. This report compares hydrophobic concrete with traditional OPC concrete, particularly in its application for slanted roofing, which is critical in preventing water accumulation and damage. The project involved designing a slanted roof where half of the surface was made with OPC concrete and the other half with hydrophobic concrete, allowing for direct comparison.

How to Make Hydrophobic Concrete (Mix Design)

Hydrophobic concrete is made by incorporating a **hydrophobic agent** such as **oleic acid** into the cement mix. In this project, we started with a mix design that closely followed the OPC mix proportions but added **oleic acid** to create the water-repellent properties.

Dimensions of the Half-Sided Roofs

The model involved a slanted roof divided into two halves:

- **Half with OPC concrete**.
- **Half with hydrophobic concrete**.

The total dimensions of the slanted roof were:

- **Length**: 46.5 cm (0.465 m)
- **Width**: 33.5 cm (0.335 m)
- **Height**: 2.5 cm (0.025 m)

For each half:

- **Length**: 0.465 m / 2 = 0.2325 m
- **Width**: 0.335 m / 2 = 0.1675 m
- **Height**: 0.025 m

Mix Design for Hydrophobic Concrete

Mix Design for Hydrophobic Concrete

For the final volume of 0.00112 m^3 with 20 grams of oleic acid added:

1. Cement:

• 500 kg/m^3 x 0.00112 m^3 = 560 grams

2. Natural Fine Sand:

• 1500 kg/m^3 × 0.00112 m^3 = 1680 grams

3. Fly Ash (20% of cement weight):

• 100 kg/m^3 x 0.00112 m^2 = 112 grams

4. Water (W/C Ratio = 0.45): 0.45 x 560 grams = 252 gram

5. Oleic Acid: 20 grams (final amount after trial and error)

Why We Chose These Specific Proportions

- **Cement**: 500 kg/m^3 was selected based on standard guidelines to ensure sufficient strength and structural integrity.

- **Fine Sand**: 1500 kg/m^3 was used for achieving good workability and proper bonding between particles.
- **Fly Ash**: 20% of cement weight was chosen to improve durability, decrease permeability, and add strength.
- **Water-Cement Ratio**: A ratio of **0.45** was chosen to reduce porosity while maintaining workability, which is critical in creating a dense, hydrophobic concrete.
- **Oleic Acid**: Through experimentation, **20 grams** of oleic acid was found to produce effective water-repellent results. We started with lower quantities (2-4 grams), but this amount was insufficient for achieving hydrophobicity.

How to Prepare Hydrophobic Concrete

1. **Dry Mixing**:
    a. Combine **cement, fly ash, and sand** in a container and mix thoroughly until the dry ingredients are uniformly distributed.
2. **Prepare Oleic Acid Solution**:
    a. Dissolve **oleic acid** in part of the water (approx. 100 grams). Stir thoroughly to ensure the oleic acid is evenly mixed with the water.
3. **Wet Mixing**:
    a. Gradually add the **oleic acid solution** to the dry mix while stirring.
    b. Add the remaining water in stages and continue mixing until a smooth and consistent mixture is achieved.
4. **Casting**:
    a. Pour the hydrophobic concrete into the mold, ensuring that it is compacted to remove any air voids.
5. **Surface Finishing**:
    a. Smooth the surface of the concrete using a trowel, ensuring an even finish.

How to Cure Hydrophobic Concrete

Curing is critical for developing the hydrophobic properties of concrete. The slab should be kept **moist** for at least **7 days** to allow for proper hydration. After casting, the slab was covered with **plastic sheeting** to prevent rapid evaporation and retain moisture. Water was applied periodically to maintain surface moisture.

Advantages of Hydrophobic Concrete

- **Water Resistance**: Hydrophobic concrete repels water, preventing moisture ingress and protecting the structure from water damage.
- **Durability**: The addition of fly ash improves durability and strength, making the concrete more resistant to cracking and chemical attacks.
- **Lower Maintenance**: Since hydrophobic concrete reduces water absorption, it minimizes the need for repairs related to water damage.
- **Energy Efficiency**: Combined with reflective coatings, hydrophobic concrete can reduce heat absorption, making it ideal for energy-efficient roofing systems.

Limitations and Challenges

1. **Challenges Faced**:
    a. **Initial trials** with lower dosages of oleic acid (2-4 grams) did not produce the expected water-repellent effect. After several adjustments, it was concluded that **20 grams** of oleic acid was required to achieve proper hydrophobicity.
2. **How We Overcame These Challenges**:
    a. **Increased Dosage of Oleic Acid**: Based on test results, we increased the oleic acid dosage from 2-4 grams to **20 grams** to ensure full hydrophobic functionality.
    b. **Proper Mixing and Curing**: Ensuring that the oleic acid was evenly dispersed, and that proper curing was maintained was essential to achieving the desired result.

Applications of Hydrophobic Concrete

- **Slanted Roofs**: As demonstrated in this project, hydrophobic concrete is ideal for slanted roofing applications to prevent water accumulation and seepage.
- **Foundations and Basements**: Hydrophobic concrete is effective in reducing water penetration in underground structures, preventing water-related damage.
- **Outdoor Pavements**: It is useful for roads, walkways, and other outdoor surfaces exposed to rain.
- **Marine Structures**: Hydrophobic concrete offers protection against saltwater corrosion, making it ideal for use in docks, piers, and bridges.

Conclusion

This project successfully demonstrated the effectiveness of **hydrophobic concrete** compared to traditional OPC concrete. Through experimentation and adjustments, we developed a mix design that met the necessary criteria for water resistance, particularly for slanted roof applications. The use of **20 grams of oleic acid** proved critical in achieving the desired hydrophobic properties. Hydrophobic concrete offers a promising solution for water-sensitive structures, reducing maintenance, increasing durability, and ensuring long-term protection against moisture ingress.

Reference: https://doi.org/10.1016/j.jobe.2024.110600